

CLASS XII · LECTURE 2

Data Types, Operators and Expressions

Option I · Two periods · A Section C question is "types of operators"

By the end of this lecture you can

- Choose int, float, double or char and the matching printf or scanf code.
- List the operator families with one example each.
- Evaluate an integer expression left to right, showing the working.
- Rewrite an algebraic formula as a legal C expression.

The types you actually use

Type	Holds	Format for printf / scanf
int	Whole numbers	%d
float	Numbers with a fractional part, single precision	%f
double	Fractional numbers with more precision	%lf in scanf, %f in printf
char	One character	%c
void	No value. Used for a function that returns nothing	none

Declare a variable before you use it: `int marks;` or `float average;`. You can initialise as you declare: `int n = 0;`. A constant that must not change can be written with `#define` or with `const int PASS = 40;`. The `%` operator, remainder, is only for integers. `10 % 3` is 1. You do not write `%` on a float.

Families of operators

A Section C answer is this table turned into sentences, plus one line of code for each family. Arithmetic operators do the calculation. Relational operators compare and give 1 or 0. Logical operators join comparisons. Assignment stores a value. Increment and decrement change a variable by one.

Family	Operators	Example and result
Arithmetic	<code>+ - * / %</code>	If a is 7, <code>a % 2</code> is 1. Integer <code>7 / 2</code> is 3, not 3.5
Relational	<code>< > <= >= == !=</code>	<code>marks >= 40</code> is 1 when the student has passed
Logical	<code>&& !</code>	<code>age >= 16 && age <= 19</code> is true only when both tests are true
Assignment	<code>= += -=</code>	<code>n += 2</code> means <code>n = n + 2</code>
Increment	<code>++ --</code>	<code>n++</code> uses n, then adds 1. <code>++n</code> adds 1, then uses n

Among `+`, `-`, `/` and `%`, the higher precedence belongs to `/` and `%`, equally. They are done before addition and subtraction, and from left to right when they sit together. `*` has that same high level. So `a + b * c` multiplies first. Use brackets whenever you are not certain. Brackets also make an exam expression readable, and readable working scores.

ON THE BOARD Post-increment

`k++` is post-increment: the current value of k is used, then k becomes `k + 1`. `++k` is pre-increment: k increases first, then the new value is used. The multiple-choice phrase "an example of a post-increment operator" is `k++`.

Integer expressions, worked the way the paper sets them

Let `a = 2`, `b = 5` and `c = 10`, all integers. Show every step. Integer division throws away the fraction.

Expression	Working	Result
$200 * ((a * 10) + c)$	$a * 10 = 20; 20 + 10 = 30; 200 * 30 = 6000$	6000
$a * b * 10 / c + a + c$	$2 * 5 = 10; 10 * 10 = 100; 100 / 10 = 10; 10 + 2 = 12; 12 + 10 = 22$	22
$a - (c / b) / 2 + 100$	$10 / 5 = 2; 2 / 2 = 1; 2 - 1 + 100 = 101$	101

Algebra into C

C has no superscripts and no implied multiplication. $3ab$ must become $3 * a * b$. A fraction needs a slash. A square is a name multiplied by itself. The absolute value needs `fabs` from `math.h`.

Mathematics	Legal C
$A = (1/2) \times \text{base} \times \text{height}$	$A = 0.5 * \text{base} * \text{height};$
$x = 3ab^3 + 3a^2b$	$x = 3 * a * b * b * b + 3 * a * a * b;$
$d = b^2 - 4ac $	$d = \text{fabs}(b * b - 4 * a * c);$
$\text{area} = \pi r^2$	$\text{area} = 3.1416 * r * r;$

EXAM TIP The half that becomes zero

If base and height are integers, writing $(1/2) * \text{base} * \text{height}$ gives 0, because $1/2$ in integer arithmetic is 0. Write 0.5, or divide by 2.0. This is the mistake that turns a correct formula into a wrong program.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. The % operator is used with:

- A) float only
- B) int
- C) a string
- D) char only

2. Which is a post-increment?

- A) ++k
- B) k++
- C) k += 1 written as a comparison
- D) k == 1

Short answers

S1. If $a = 2$, $b = 5$ and $c = 10$, all int, find the value of $a * b * 10 / c + a + c$. Show the working.

S2. Convert these into C: $A = (1/2) \times \text{base} \times \text{height}$, and $d = |b^2 - 4ac|$.

Detailed answer

L1. What are the different types of operators used in a C program? Explain with examples.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. B. Remainder is defined for integers.

MCQ 2. B. The plus signs come after the variable.

S1. Work left to right for * and /. $2 * 5 = 10$. $10 * 10 = 100$. $100 / 10 = 10$. Then $10 + a + c = 10 + 2 + 10 = 22$. The result is 22.

S2. $A = 0.5 * \text{base} * \text{height}$; and, after `#include <math.h>`, $d = \text{fabs}(b * b - 4 * a * c)$; Do not write $1/2$ as an integer division if you need a half, because $1/2$ is 0 in integer arithmetic.

L1. An operator is a symbol that asks C to do something to its operands. Arithmetic operators are + − * / and %. If n is 7, $n \% 2$ is 1, and integer $7/2$ is 3. Relational operators are < > <= >= == and !=. They compare two values and give 1 or 0, as in marks >= 40. Logical operators are &&, || and !. They join conditions: $\text{age} \geq 16 \ \&\& \ \text{age} \leq 19$. Assignment operators store a result: $n = 5$ and $n += 2$. Increment and decrement are ++ and --. k++ is post-increment. / and % are done before + and −. Brackets force the order you want. The % operator is only for integers.